

Syllabus for “Econometrics I”

Winter 2025-26

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Chair of Economic Growth and Development

MLU Halle-Wittenberg, Universitätsring 3

Time and location:

- Tuesday: 10:15 - 11:45, weekly (from 14/10/25), Hörsaal A [Mel]*
- Thursday: 14:15 - 15:45, weekly (from 16/10/25), Hörsaal XX [Mel]*
- Thursday: 16:15 - 17:45, weekly (from 16/10/25), Hörsaal B [Mel]*
- Monday: 16:15 - 17:45, weekly (from 20/10/25), Großer Hörsaal [WiWi]

* “Mel” stands for *Melanchthonianum*, a building located in Universitätsplatz 9.

We will start on Tuesday, the **14th of October** (note that our first Monday meeting takes place the second week). There is no sharp distinction between lectures and tutorials: we will have a mixture of both, spending roughly half of the time on tutorial-like exercises.

Contents:

- Multivariate OLS estimator: Coefficients, standard errors, significance, predictions, properties, asymptotic behavior.
- What variables should be included in a regression? Omitted variable biases, efficiency-bias tradeoff.
- Departures from the classical model: Heteroscedasticity, endogeneity, common misspecifications.
- Basic applications: Dummies, fixed effects, binary dependent variables, introduction to time series and panel data.
- From correlation to causality: Basic methods to distinguish between them (experiments and quasi-experiments, difference in difference, instrumental variables).
- Hands-on analysis: handling databases, creating graphs, making simulations, running regressions, and producing result tables. We will use the free econometric software *R* (<http://www.r-project.org/>) and the user interface *Rstudio* (<https://rstudio.com/products/rstudio/download/>).

Examination:

Written examination on December 1, 2025 and February 23, 2026 (Großer Hörsaal [WiWi]).
The exact date and location can be changed by the Examinations Office.

Literature:

- Hill, R., Griffiths, W. & Li, G. (2018) “*Principles of Econometrics*,” 5th edition (available in library; other editions can be used too)
- Wooldridge, J. (2020) “*Introductory Econometrics. A Modern Approach*,” 7th edition (available in library; other editions can be used too)
- Greene, W. (2019) “*Econometric Analysis*,” 8th edition (available in library; other editions can be used too)
- Heiss, F. (2020) “*Introductory Econometrics using R*” (available online: <http://www.urfie.net/>)